EXPLAINABLE - DRIFT-AWARE - COST-ACCOUNTED

TabML-Guard Report — Metro Interstate Traffic Volume

data/metro_traffic.csv · 48,204 rows × 7 features · regression

RECOMMENDED MODEL

catboost_default

RMSE (HELD-OUT)

1893.9764

AT A GLANCE

- Defaults beat tuning
- Inputs drifted, performance held

Headline finding

catboost_default is the recommended model, scoring 1893.9764 rmse on a strictly held-out, time-ordered test set.

Recommendation

No search beat the library defaults on the held-out period, so the default configuration is the pragmatic production choice. At an equal budget Bayesian search did not beat random search - on a space this size, random is the pragmatic default. The input distribution shifted materially (PSI above 0.25) while accuracy held up - monitor, but no retrain is urgent yet.

This report was produced automatically by the TabML-Guard pipeline. Every number below comes from a single run on a strictly held-out, time-ordered test split - no test data was used for model selection or tuning.

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STAGE 0

Dataset, profile and configuration

The pipeline begins by profiling the raw table and deciding, from the target column alone, whether this is a classification or a regression problem. When a time column is present the split is chronological - the model trains on the past and is scored on the future.

Rows	48,204
Features	7
Target column	traffic_volume
Detected task	regression
Missing values	48,143
Time column	date_time
Time span	2012-10-02 09:00:00 -> 2018-09-30 23:00:00
Target statistics	min=0.0000 max=7,280 mean=3259.8184 std=1986.8607

Configuration used

This is the entire dataset-specific input to the run. Everything else is generic pipeline code.

```
dataset.source        csv
dataset.name          -
dataset.path          data/metro_traffic.csv
dataset.target        traffic_volume
dataset.time_col      date_time
dataset.drop_cols     []
split                 train=0.6  test=0.4
models.boosters       ['xgboost', 'lightgbm', 'catboost', 'random_forest']
models.stacking       True
tuning.budget_trials  40
tuning.cv_splits      3
drift.n_chunks        10
drift.retrain         sliding
random_state          42
```

STAGE 1

Regularized linear baselines

Before reaching for an ensemble, the pipeline establishes what a well-regularized linear model can do. This is the honesty check: if a linear model is competitive, the extra complexity of boosting has to justify itself.

model	rmse	mae	r2	fit
elasticnet	**1921.5313**	**1663.7564**	**0.0542**	**13.6s**
ridge	1921.9234	1663.1962	0.0538	0.1s
linear	1922.1854	1663.3442	0.0536	0.1s
lasso	1922.4679	1663.6930	0.0533	720.2s

the 4 regularizers span only 0.9366 rmse (best 'elasticnet' 1921.5313) — with abundant rows and few features, penalties barely move the fit at the default strength; Lasso path: at alpha=10 only 10/61 features survive (sparsity)

STAGE 2

Ensemble and boosting models

Gradient-boosted trees plus a stacking meta-model, all scored through the same evaluation path as the baselines so the leaderboard rows are directly comparable.

model	rmse	mae	r2	fit
catboost	**1894.3917**	**1633.3311**	**0.0807**	**1.5s**
lightgbm	1909.6062	1635.9541	0.0659	0.4s
stacking	1911.9557	1637.8531	0.0636	17.0s
xgboost	1921.5102	1641.2971	0.0542	0.5s
elasticnet	1921.5313	1663.7564	0.0542	13.6s
ridge	1921.9234	1663.1962	0.0538	0.1s
linear	1922.1854	1663.3442	0.0536	0.1s
lasso	1922.4679	1663.6930	0.0533	720.2s
random_forest	2181.7100	1784.9568	-0.2192	2.9s

Best ensemble: **catboost**. The strongest single booster (**catboost**) is the one carried into the tuning comparison in Stage 3.

STAGE 3

Hyperparameter optimization: benefit vs cost

Four strategies compete on **catboost**: library defaults, an exhaustive coarse grid, random search, and Bayesian optimization (Optuna TPE). Random and Bayesian are given an **equal budget** of 40 trials, which makes them a fair head-to-head. Crucially, the table reports not just the score but what each strategy *cost*.

strategy	test rmse	wall clock	model fits	cv score
default	**1893.9764**	**1.5s**	**1**	**-**
grid	1906.0665	11.9s	54	-1929.2632
random	1901.9605	26.2s	120	-1929.8724
bayes	1906.8466	105.9s	120	-1929.4687

The *cv score* column is the internal cross-validated score each search optimized (scikit-learn's convention is higher-is-better, so error metrics appear negated). The *test* column is the winning configuration refit once and scored on the untouched held-out period - that is the number to trust.

> **Is tuning worth it?**
>
> tuning did NOT beat the default (rmse=1893.9764); on this data a sensible default is already competitive; at equal budget, random search won (rmse: bayes 1906.8466 @ 105.92s vs random 1901.9605 @ 26.2s)

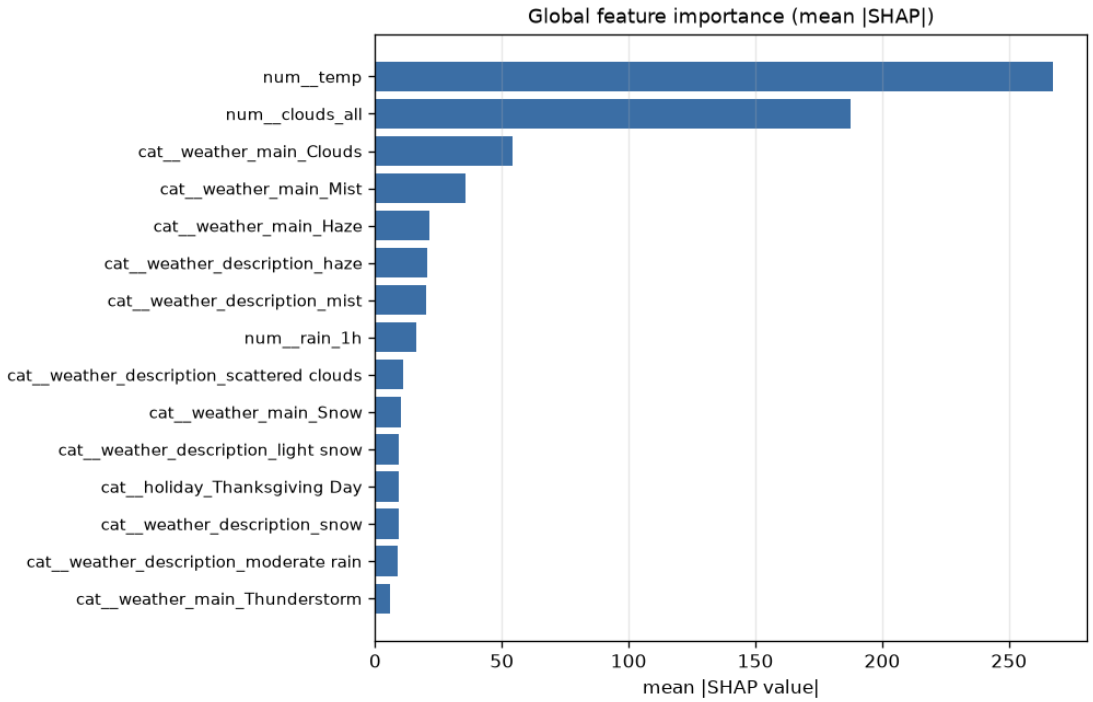
STAGE 4

Explainability and prediction reliability

SHAP attributions for **catboost_default**, using the explainer matched to the model family (tree, linear, or model-agnostic fallback). Global importance answers *what drives this model overall*; the local example shows how a

single prediction was reached.

feature	mean |SHAP|
num__temp	267.51273
num__clouds_all	187.37917
cat__weather_main_Clouds	54.16446
cat__weather_main_Mist	35.82227
cat__weather_main_Haze	21.25925
cat__weather_description_haze	20.59202
cat__weather_description_mist	20.11540
num__rain_1h	16.15344
cat__weather_description_scattered clouds	10.88241
cat__weather_main_Snow	10.31056
cat__weather_description_light snow	9.53003
cat__holiday_Thanksgiving Day	9.31086
cat__weather_description_snow	9.27895
cat__weather_description_moderate rain	9.10256
cat__weather_main_Thunderstorm	5.89093



Global feature importance (mean |SHAP|)

reliability analysis unavailable.

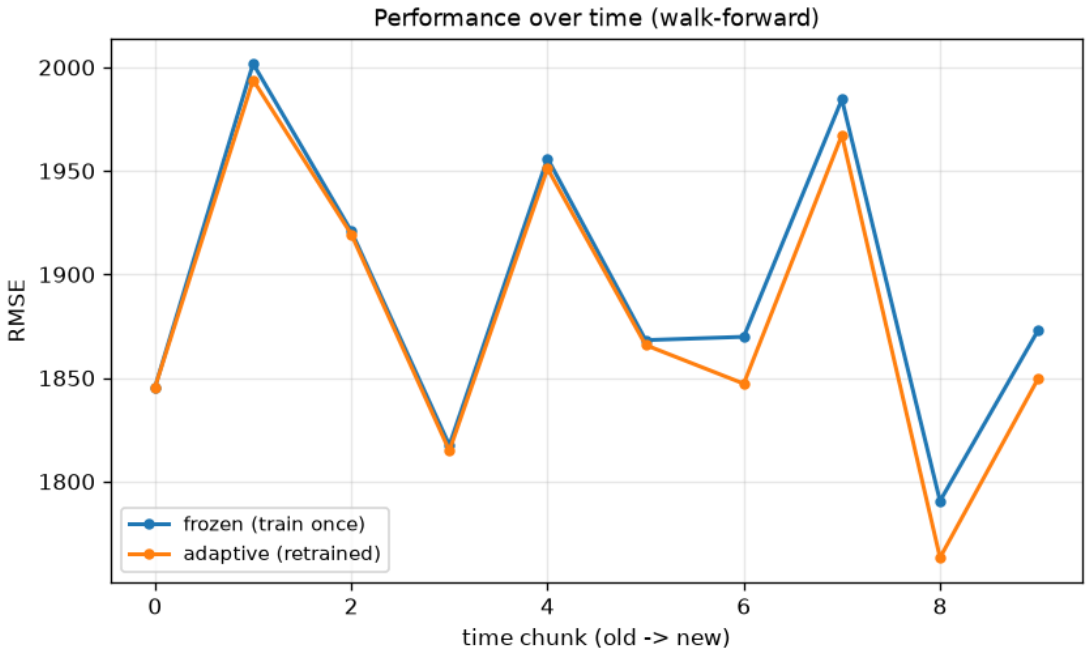
STAGE 5

Concept drift and statistical significance

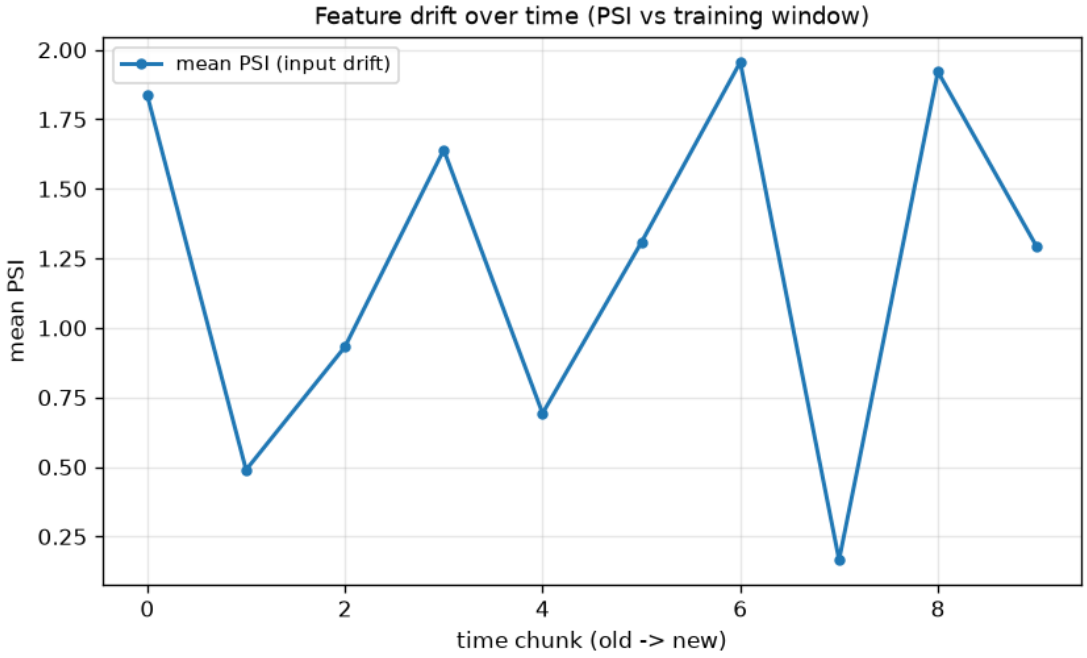
The held-out period is replayed in 10 chronological chunks. A **frozen** model (trained once, never updated) is compared against an **adaptive** model that predicts each chunk *before* learning from it - so there is no look-ahead. Retraining mode: **sliding**.

chunk	rows	frozen	adaptive	mean PSI
0	1,929	1845.4785	1845.4785	1.837
1	1,929	2001.7474	1993.8780	0.490
2	1,928	1920.8953	1919.4334	0.932
3	1,928	1817.4988	1814.8576	1.640
4	1,928	1956.0219	1951.1330	0.693
5	1,928	1868.2530	1865.8851	1.308
6	1,928	1869.8608	1847.2208	1.955
7	1,928	1984.4940	1967.2481	0.166
8	1,928	1790.5649	1763.2109	1.921
9	1,928	1873.0475	1849.7095	1.292

Chunk 1 is a built-in consistency check: both models are identical there by construction, since neither has seen the chunk yet.



Frozen vs adaptive performance across the held-out period



Input drift (PSI) against the training window

Mean score, frozen model	`1892.7862`
Mean score, adaptive model	`1881.8055`
First third -> last third	`-  ->  -`
Performance worsened	`no`
Worst-drifting feature	`-`
Peak mean PSI	`1.955`

Drift verdict

Frozen-model RMSE held up across time (1845.478 -> 1873.047; drift magnitude -40.005). Peak input drift (mean PSI) = 1.955 (notable). A sliding-retrained model beat the frozen one (RMSE 1881.805 vs 1892.786, gap 10.981 and the difference is statistically significant). Champion vs linear baseline: significant (p=9.26626e-28).

Statistical significance

A score difference is only meaningful if it survives a significance test. Paired predictions on the same held-out rows are compared with the Wilcoxon signed-rank test. A p-value below 0.05 means the difference is unlikely to be chance; above it, the two models are statistically indistinguishable - which is itself worth reporting.

comparison	better model	p-value	conclusion
champion vs runner up	catboost	`7.582e-06`	`significant`
champion vs linear baseline	catboost_default	`9.266e-28`	`significant`
static vs adaptive	adaptive	`5.004e-18`	`significant`

champion vs runner up: catboost has lower absolute error; the difference is statistically significant (p=7.582e-06).

champion vs linear baseline: catboost_default has lower absolute error; the difference is statistically significant (p=9.266e-28).

static vs adaptive: adaptive has lower absolute error; the difference is statistically significant (p=5.004e-18).

APPENDIX A

Scope and limitations

Tuning is measured, not assumed

The point of Stage 3 is to quantify the cost/benefit of tuning, not to prove that tuning always wins. On well-conditioned datasets a regularized baseline can match a tuned booster, and this report says so when it happens.

Drift is detected, not explained

PSI and prequential error are cheap, reliable indicators that a distribution has moved. Neither attributes a cause; deeper causal drift analysis is out of scope.

Single-machine scale

The pipeline targets datasets that fit in memory on one machine. It is not a distributed training system.

One split, one seed

Results come from a single temporal split at a fixed random seed. That is the right choice for a time-ordered problem - shuffled k-fold would leak the future - but it means confidence intervals across resamples are not reported, which is why the significance tests above matter.

APPENDIX B

Glossary

ROC-AUC	Probability a random positive is ranked above a random negative. 0.5 is chance, 1.0 is perfect. Insensitive to class imbalance.
RMSE / MAE	Average prediction error in the target's own units. Lower is better; RMSE punishes large misses harder than MAE.
R-squared	Share of the target's variance the model explains. 0 means no better than predicting the mean.
Regularization (L1/L2)	A penalty on coefficient size that trades a little bias for less variance. L1 can zero coefficients outright, performing feature selection; L2 shrinks them smoothly.
Stacking	Training a small meta-model on the base models' predictions so it can learn which to trust where.
Bayesian optimization	Tuning that builds a surrogate model of the search space to choose each next trial, rather than sampling blindly.
SHAP value	A feature's signed contribution to one prediction, derived from cooperative game theory. Contributions sum to the prediction.
PSI	Population Stability Index: how far a feature's distribution has moved from its training reference. Rules of thumb: below 0.1 stable, 0.1-0.25 moderate, above 0.25 material.
Prequential evaluation	Test-then-train: each chunk is predicted before the model is allowed to learn from it, so no future information leaks.
McNemar's test	A paired test for two classifiers on identical rows, based only on the cases where they disagree.

Produced by the TabML-Guard pipeline.

~ Ayush Gole